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T. B. C. : GPGM – 2/20

Test Booklet Series

A

TEST BOOKLET

(GEOLOGIST)

PAPER – II

21277

Sl. No.

Time Allowed : 3 Hours

Maximum Marks : 300

: INSTRUCTIONS TO CANDIDATES :

1. IMMEDIATELY AFTER THE COMMENCEMENT OF THE EXAMINATION, YOU SHOULD CHECK THAT THIS TEST BOOKLET **DOES NOT** HAVE ANY UNPRINTED OR TORN OR MISSING PAGES OR ITEMS ETC. IF SO, GET IT REPLACED BY A COMPLETE TEST BOOKLET OF THE SAME SERIES ISSUED TO YOU.
2. ENCODE CLEARLY THE TEST BOOKLET SERIES **A, B, C** OR **D**, AS THE CASE MAY BE, IN THE APPROPRIATE PLACE IN THE ANSWER SHEET USING BALL POINT PEN (BLUE OR BLACK).
3. You have to enter your **Roll No.** on the Test Booklet in the Box provided alongside. **DO NOT** write *anything else* on the Test Booklet.
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5. This Test Booklet contains **150** items (questions). Each item (question) comprises four responses (answers). You have to select the correct response (answer) which you want to mark (darken) on the Answer Sheet. In case, you feel that there is more than one correct response (answer), you should mark (darken) the response (answer) which you consider the best. In any case, choose **ONLY ONE** response (answer) for each item (question).
6. You have to mark (darken) all your responses (answers) **ONLY** on the **separate Answer Sheet** provided by **using BALL POINT PEN (BLUE OR BLACK)**. See instructions in the Answer Sheet.
7. All items (questions) carry equal marks. All items (questions) are compulsory. Your total marks will depend only on the number of correct responses (answers) marked by you in the Answer Sheet. **There will be no negative marking for wrong response (answer).**
8. Before you proceed to mark (darken) in the Answer Sheet the responses (answers) to various items (questions) in the Test Booklet, you have to fill in some particulars in the Answer Sheet as per the instructions sent to you with your **Admission Certificate**.
9. After you have completed filling in all your responses (answers) on the Answer Sheet and after conclusion of the examination, you should hand over to the Invigilator the *Answer Sheet* issued to you. You are allowed to take with you the candidate's copy / second page of the Answer Sheet along with the **Test Booklet**, after completion of the examination, for your reference.
10. Sheets for rough work are appended in the Test Booklet at the end.

SEAL

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1. Residual hill in the desert is called :

(A) Inselberg

(B) Playa

(C) Padiment

(D) Inlier

2. A process of chemical weathering is due to :

(A) Mass exfoliation

(B) Hydration and hydrolysis

(C) Frost action

(D) Thermal expansion and contraction

3. Deflection of wind due to the rotation of the Earth is called _____.

(A) Polar wind

(B) Geostrophic wind

(C) Easterlies

(D) None of these

4. The highest Plateau in the world is :

(A) Colorado Plateau

(B) Pamir Plateau

(C) Patagonia Plateau

(D) Potwar Plateau

5. The intensity of chemical weathering is more in :

(A) Permafroast region

(B) Hot and dry region

(C) Hot and humid region

(D) Cold climatic region

6. Broad slope of debris deposited by the coalescing of several alluvial fans along the lower slopes of mountains by descending steams is called :

(A) Alluvial deposits

(B) Colluvial deposits

(C) Stream deposits

(D) Bajada

7. The geomorphology of river Brahmaputra in Assam is :

(A) Braided river

(B) Influent river

(C) Affluent river

(D) None of these

8. The bowl-shaped, amphitheater-like depressions that glaciers carve into mountains at high elevations is known as :

(A) Nunataks, aretes and horns

(B) Moraines

(C) Fjords and handing valley

(D) Cirques

9. It can be expressed the amount of runoff in running water cycle by the relation : (fill in the blank selecting appropriate term) : Runoff= precipitation – (? + Evaporation + Transpiration)

(A) Discharge

(B) Turbulent flow

(C) Infiltration

(D) None of these

10. An **alluvial fan** is the land counterpart of a _____.

(A) Delta

(B) Valley floor

(C) Yazoo

(D) Alluvial fan

11. Volcanic landforms tend to be **cinder cones, stratovolcanoes, _____ and volcanic domes.**

Choose the missing volcanic landform :

(A) Fold

(B) Dunes and Plateaus

(C) Canyon

(D) Calderas

12. Karst topography is a type of landform found in _____.

(A) Limestone country

(B) Sandstone dominated areas

(C) Basaltic domain

(D) Bauxite dominated regime

13. There are **three** main ways that scale is indicated on a toposheet : graphic (or bar), _____ and Representative Fraction (RF).

Find out the missing term :

(A) Saddle

(B) Verbal

(C) Legend and Keys

(D) Triangulated stations

14. Which one of the following is not a glacio-fluvial depositional feature ?

(A) Horn

(B) Drumlin

(C) Kame

(D) Esker

15. Wind erosion may be carried out in the following ways :

(I) Deflation

(II) Abrasion

(III) Attrition

Choose the correct option from the following codes :

(A) (I) and (III)

(B) (I) and (II)

(C) (I), (II) and (III)

(D) (II) and (III)

16. Migmatite is associated with which type of fold ?

(A) Suprataneous fold

(B) Reclined fold

(C) Ptygmatic fold

(D) Open fold

17. Within a three-dimensional solid body, how many components are required to define stress sytem at a point ?

(A) Nine

(B) Four

(C) Three

(D) Five

18. The high angle relationship between bedding and cleavage is common in :

(A) Fold limb

(B) Fold hinge

(C) Overturned fold

(D) Chevron fold

19. Transform fault is a :

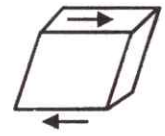
(A) Normal fault

(B) Reverse fault

(C) Thrust fault

(D) Strike-slip fault

20. What kind of stress the figure shows ?



(A) Tensional stress

(B) Extensional stress

(C) Compressional stress

(D) Shear stress

21. Salt dome is an example of :

(A) Diapiric fold

(B) Reclined fold

(C) Drag fold

(D) Overturned fold

22. The line of maximum curvature in a fold is termed as :

(A) Hinge

(B) Axis

(C) Trough

(D) Crest

23. When the hanging wall is moved down relative to the foot wall, the fault is known as :

(A) Reverse

(B) Normal

(C) Strike-slip

(D) Dip-slip

24. Younger beds will always be found in the :

(A) Reverse direction of the dip

(B) Direction of strike line

(C) Direction of dip

(D) Inclined direction of dip and strike

25. The tightness of a fold may be estimated in sections :

(A) Parallel to the fold axis or the axial plane

(B) Parallel to fold axis and perpendicular to the axial plane

(C) Perpendicular to the profile plane of the fold

(D) Perpendicular to the fold axis

26. Maximum dip of a plane can be obtained on a line that is at _____ angle with strike of the bed.

(A) 25°

(B) 45°

(C) 90°

(D) 65°

27. If the strike of the dipping bed is $N25^{\circ}E$, the dip direction is :

(A) $S65^{\circ}W$

(B) $S65^{\circ}E$

(C) $S25^{\circ}W$

(D) $S25^{\circ}E$

28. The two sides of a fold are named as :

(A) Axial planes

(B) Anticlines

(C) Hinges

(D) Limbs

29. "In elastic deformation, strain is proportional to stress" is related to :

(A) Hooke's law

(B) Smith's law

(C) Poisson's ratio

(D) Modulus of elasticity

30. The line of maximum curvature in a fold is known as :

(A) Crest

(B) Hinge

(C) Axis

(D) Trough

31. The Volcanoes of Galapagos Islands are of interest because of their extraordinary zoning and relationship to a nearby _____.

(A) Spreading ridge

(B) Subduction centre

(C) Transform boundary

(D) Volcanic process

32. Hot mantle from the two adjacent cells rises at the _____ creating new ocean crust.

(A) Deep sea trench

(B) Volcanic arcs

(C) Mid oceanic ridge axis

(D) Subduction zones

33. The Indian Peninsula was a part of _____.

(A) Eurasia

(B) Laurasia

(C) Gondwana

(D) All of these

34. With the separation from mid oceanic ridge via wedening of sea floor, an important phenomenon is recorded in magmatic rocks is known as _____.

(A) Static record of electric reversal

(B) Magnetic polarity reversal

(C) Thermal reversal

(D) All of these

35. The theory of Alfred Wegner is about :

(A) Magnetic reversal

(B) Sea-floor spreading

(C) Continental drift

(D) Isostasy

36. The Pacific Ocean where many volcanic eruptions and earthquakes occur is known as :

(A) Pacific Ring of Fire

(B) Girdle of Fire

(C) Ring of Fire

(D) All of these

37. Which of the following doesn't represent a plate boundary ?

(A) Convergent

(B) Divergent

(C) Transform

(D) Subduction

38. The old crust melte back into the formation of magmatic fluid along :

(A) Mid ocean ridges

(B) Sea shore

(C) Volcanic islands

(D) Convergent plate boundaries

39. The Sierra Nevada in California, the Harz Mountains in Germany are the examples of :

(A) Fault-block mountains

(B) Fold mountains

(C) Plateau mountains

(D) Dome mountains

40. An accretionary prism forms from sediments accreted onto the non-subducting tectonic plate at a :

(A) Convergent plate boundary where an oceanic plate collides with a continental plate

(B) Transform plate boundary

(C) Divergent plate boundary

(D) All of these

41. Name the Geologist who proposed the term "Isostasy" :

(A) C. Dutton

(B) Alfred Wegner

(C) Harold Hammond Hess

(D) Robert Dietz

42. What is the density of the earth in the core immediately after the mantle ?

(A) 8.8 g/cm^3

(B) 8.9 g/cm^3

(C) 9.9 g/cm^3

(D) 12.0 g/cm^3

43. Features that are not found in convergent plate boundary :

(A) Deep ocean trenches

(B) Island arcs

(C) Submarine valley

(D) Volcanoes

44. Which one of the following is associated with a mid ocean ridge ?

(A) Basaltic pillow lava

(B) Ocean trench

(C) Deep focus earthquakes

(D) All of these

45. The discontinuity between outer and inner core is known as :

(A) Mohorovicic discontinuity

(B) Gutenberg discontinuity

(C) Lehmann discontinuity

(D) Repetti discontinuity

46. Arrange the following formations of the Gondwana Supergroup in order of their decreasing age (Choose the correct option) :

(I) Raniganj Formation

(II) Barakar Formation

(III) Kulti Formation

(IV) Karharbari Formation

(A) (I), (III), (IV), (II)

(B) (IV), (I), (II), (III)

(C) (II), (I), (IV), (III)

(D) (IV), (II), (III), (I)

47. Which of the following is the type area of Triassic rocks in India ?

(A) Kutch

(B) Spiti

(C) Trichinappally

(D) Jaisalmer

48. The age of Bagh Beds is :

(A) Upper Cretaceous

(B) Lower Jurassic

(C) Triassic

(D) Upper Permian

49. The age of Fenestella Shales, India is :

(A) Ordovician

(B) Permian

(C) Carboniferous

(D) Devonian

50. The Deccan Traps occur mainly in Maharashtra. Other states include a part of (identify the state where Deccan Trap doesn't occur) :

(A) Karnataka

(B) Orissa

(C) Madhya Pradesh

(D) Gujrat

51. Age of the oldest rocks in the ocean basin is :

(A) Jurassic

(B) Permian

(C) Devonian

(D) Cambrian

52. The Proterozoic Delhi Supergroup is classified into unconformity-based three groups (identify the odd one) :

(A) Raialo

(B) Ajabgarh

(C) Aravalli

(D) Alwar

53. Choose the correct statement from the following :

(A) Limestone is not found in Vindhayan Supergroup

(B) The top of the Cuddapah Supergroup is Phanerozoic

(C) Kaladgi basin is located in the Baster Craton

(D) Sandstone is dominant in the Mesozoic Bhuj Formation

54. Sedimentary cycles often display presistent trends in thickness and facies composition, called stacking patterns, and these fall into which types ?

(A) Retrogradational

(B) Aggradational

(C) Progradational and degradational

(D) All of these

55. The concept of 'uniformitarianism was propunded by :

(A) Hutton

(B) Wegner

(C) Dutton

(D) Kant

56. Trilobites and graptolites occur abundantly during _____.

(A) Cambrian and Ordovician

(B) Devonian and Carboniferous

(C) Proterozoic

(D) Triassic and Jurassic

57. The following are related to the Cuddapah Basin, Identify which is not related to the Basin :

(A) Crescent shaped foreland basin in the frong of EGMB

(B) Proterozoic sedimentary rocks rest up against Archean gneisses and granulites

(C) The sedimentary pile are highly metamorphed

(D) Forms Nallamali Range parallel to the Eastern Ghats

58. The Panjal Traps of Kashmir show significant outpouring of mafic and silicic volcanics during the Early Permian and are synchronous with the opening of the _____

(A) Proto-Mediterranean Ocean

(B) Baltica Ocean

(C) Indian Ocean

(D) Neotethys Ocean

59. Lameta Formation occuring in different Indian States hold fossils of Cretaceous sauropod dinosaur. Identify the state where dinosaur is found in other formation :

(A) Madhya Pradesh

(B) Gujarat

(C) Maharashtra

(D) Tamil Nadu

60. Nimar Sandstone of Cretaceous age belongs to _____.

(A) Belemnite Beds

(B) Bagh Beds

(C) Uttattur Beds

(D) All of these

61. The earliest known fossil vertebrates discovered in rocks from the Ordovician Period about 500 to 430 Ma was _____.

(A) Dinosaur

(B) Crocodile

(C) Armored fish

(D) Lizard

62. Identify the body feature that doesn't characterize a molluscan body :

(A) Hard and segmented body

(B) Haemocoel body cavity

(C) Respiration via ctenidium

(D) Ganglia/nerve comprised nervous system

63. One of the plate of the apical disc in an echinoid shell is perforated, this plate is known as :

(A) Ocular plates

(B) Genital plate

(C) Med reproach plate

(D) Ambulacra plate

64. Macedon in an echinoid is associated with :

(A) Apical

(B) Tubercle

(C) Corona

(D) Periscope

65. The earliest fossils of foraminifera are found in :

(A) Permian rocks

(B) Carboniferous rocks

(C) Upper Cambrian rocks

(D) Silurian rocks

66. Skeletal remains of an armoured dinosaur Ornithischia : Ankylosauria is :

(A) Maleri Formation

(B) Pachmari Beds

(C) Kota Formation

(D) Bagra Formation

67. The Gondwana Flora represents the period :

(A) Upper Carboniferous to Jurassic

(B) Upper Cambrian to Jurassic

(C) Triassic and Jurassic

(D) Upper Permian only

68. Brachiopods were abundant during Palaeozoic times. First appearance of Brachiopods was reported from ;

(A) Ordovician

(B) Permian

(C) Early Cambrian

(D) Devonian

69. Mark the feature that is not related to the fossil 'Unio (bivalve)' :

(A) Medium sized saline water mussels

(B) Aquatic bivalve

(C) Belongs to the family Unionidae

(D) Unio is a genus

70. Dedoxylon jamudhiense is a new species of Fossil Wood from the Raniganj Stage of :

(A) Jharia Coalfield, Bihar

(B) Asansol Coalfield, West Bengal

(C) Salt Range proper

(D) Belemnite beds of Baluchistan

71. 'Benthic foraminifera' as an indicator of potential _____.

(A) Cold climatic condition

(B) Environmental indicator

(C) Petroleum sources

(D) All of these

72. Foraminifera are single-celled organisms (protists) often used as a/an :

(A) Indicator of paleo-environment

(B) To map past distributions of the tropics

(C) Locate ancient shorelines

(D) All of these

73. **Tribobite,** an extinct fossil arthropods is important due to the following (identify the odd one not related to Trilobite) :

(A) Recognized by distinctive three-lobbed and three-segmented form

(B) Exclusively marine animals

(C) First appeared in Cambrian period

(D) The fossil survived till Jurassic period

74. Name the fossil having the following properties :

(I) A large lenticular fossil

(II) Characterized by its numerous coils

(III) Coils are subdivided by septa into chambers

(A) Ostracod

(B) Nummulite

(C) Gastropod

(D) None of these

75. The five stages of human evolution (from primitive to the early human) are :

(I) Australopithecus

(II) Dryopithecus

(III) Homo Erectus

(IV) Homo Sapiens Neander-thalensis

(V) Ramapithecus

The correct sequence is :

(A) (II), (V), (I), (III), (IV)

(B) (II), (I), (IV), (III), (V)

(C) (IV), (I), (III), (II), (V)

(D) (II), (IV), (III), (I), (V)

76. The light-colored igneous rocks are called :

(A) Felsic

(B) Mafic

(C) Melanocratic

(D) All of these

77. The mineral content in an igneous rock is termed as :

(A) Essential minerals

(B) Accessory minerals

(C) Gangue minerals

(D) Auxillary minerals

78. Differentiation of tholeiitic magmas can be explained by fractionation of plagioclase, _____, _____, and titaniferous magnetite. Identify the pair of minerals that are essentially involved in the process.

(A) Olivine and garnet

(B) Augite and orthoclase

(C) Microcline and pyroxene

(D) Olivine and augite

79. The basalts of intra-plate volcanoes, collectively referred to as :

(A) Ocean-island basalt

(B) Within plate basalt

(C) Pillow basalt

(D) All of these

80. Basic sills of more alkaline composition are in many instances similar to _____.

(A) Aplites

(B) Komatites

(C) Tholeiitic dolerites

(D) Migmatite

81. In Rayleigh fractionation, and partitioning of trace elements under condition is governed by the equation :

(A) $C_1/C_o = F^{(D-1)}$

(B) $C_o/C_L = F^{(D-1)}$

(C) $C_o/C_L = F^{(1-D)}$

(D) None of these

Where, F = fraction of liquid remaining, D = Bulk distribution coefficient, C_o and C_L are the concentration of trace element in the liquid and in the total original assemblage respectively.

82. When large phenocrysts are set in a finer matrix, the granitic texture is known as :

(A) Hyalophitic texture

(B) Poikilitic texture

(C) Vitrophyric texture

(D) Porphyritic texture

83. The Phase Rule of J. Williard Gibbs is usually written as :

(A) F = C – P – 2

(B) F = C – P + 2

(C) F = C + P – 2

(D) None of these

84. In the Indian shield, the anorthosite complex within the _____ granulite belts, although, occur within the Chotanagpur granulite migmatite belt and Sargur belt of Karnataka.

(A) Vindhayan Supergroup

(B) Cuddapah Supergroup

(C) Eastern Ghats Granulite Belt

(D) Western Ghat Migmatites

85. Ophiolites carry large sections of which rocks ?

(A) Oceanic rocks

(B) Continental rocks

(C) Large sections of mantle rocks

(D) Peridotites

86. Proportion of plagioclase to alkali feldspar increase from 65-90% in _____ but in tonalite plagioclase content is extremely high about 90-100%. Choose the correct option.

(A) Granite

(B) Granodiorite

(C) Monzonite

(D) Olivine granite

87. Which common rock type contains diamond if it passes through diamond zone of the mantle and preserves it ?

(A) Kimberlite and Lamprophyre

(B) Only Kimberlite

(C) Only Lamprophyre

(D) Neither Kimberlite nor Lamprophyre

88. Alkaline rocks of Samchampi-Samteran are located in which state of Indian territory ?

(A) Ri Bhoi district of Meghalaya

(B) Karbi Anglong, Assam

(C) Manipur

(D) Tripura

89. The Sukinda valley in Cuttack and Jajpur districts is known for about 97% of which deposits ?

(A) Chromite with Manganese

(B) Chromite with Gold

(C) Iron ore

(D) Chromite

90. The common units for fugacity calculations are :

(A) mol m^{-2}

(B) mol m^{-3}

(C) mol m^{-1}

(D) None of these

91. Burial diagenetic processes convert siliciclastic sediments to _____ e.g., conglomerate, sandstone, shale (pick up appropriate term).

(A) Lithified sedimentary rocks

(B) Chemical sedimentary rocks

(C) Bio-chemical sediments

(D) None of these

92. The term **provenance** refers to all aspects of the :

(A) Weathering process for sediment derivation

(B) Sources of sediment supply

(C) Basin architecture

(D) Processes of sediment transport

93. Detrital sedimentary rocks represent material eroded from the Earth is also known as :

(A) Colluviums

(B) Allogenic sediments

(C) Fluvial sediments

(D) Terrigenous sediments

94. Sole marks, found on the bottom surfaces of beds. The most common form of sole mark is a _____.

(A) Flute cast

(B) Ripple marks

(C) Cross beds

(D) Mud cracks

95. Match the following :

Sediment size		Rock name
(a)	> 2 mm	(i) Sandstone
(b)	< 1/16 – 1/256 mm	(ii) Silt stone
(c)	1/16 – 2 mm	(iii) Shale/clay
(d)	< 1/256 mm	(iv) Conglomerate

Select the correct order (options are) :

(A) (a) (ii), (b) (iii), (c) (i), (d) (iv)

(B) (a) (iv), (b) (ii), (c) (i), (d) (iii)

(C) (a) (iv), (b) (iii), (c) (i), (d) (ii)

(D) (a) (iv), (b) (i), (c) (ii), (d) (iii)

96. **Stromatolites** are one of the most common trace fossils of _____.

(A) Lower Gondwana sedimentaries

(B) Tertiary limestones

(C) Archean carbonates

(D) Late Precambrian and early Paleozoic carbonates

97. The common characteristics of **Arkose Sandstone** are :

(A) Mineral composition : quartz and feldspar

(B) Immature, medium-grained (0.06 – 2 mm)

(C) Reddish brown coloured

(D) All of these

98. Bedding that is markedly trough-shaped or scoop-shaped has also been referred to as :

(A) Tabular cross-bedding

(B) Trough cross-bedding

(C) Festoon bedding

(D) Graded bedding

99. Shales containing more than about 15 percent iron oxides are called :

(A) Phosphatic shales

(B) Ferruginous shales

(C) Calcareous shales

(D) All of these

100. In glacial environment, how Eskers are defined ?

(A) Small mound-shaped accumulations of sand or gravel that form in pockets or crevasses in the ice

(B) Narrow, sinuous ridges of sediment oriented parallel to the direction of glacial advance

(C) Accumulations deposited as terraces along the margins of valley glaciers

(D) All of these

101. Both dolomite and calcite are uniaxial negative minerals, commonly occur together. They may be difficult to distinguish optically but dolomite can be readily distinguished by :

(A) ICP-MS

(B) LA-ICP-MS

(C) XRD

(D) XRF

102. Identify the important feature/s of a coastal lagoon :

(A) A shallow stretch of seawater such as a sound, channel, bay

(B) Saltwater lake near or communicating with the sea

(C) Partly or completely separated from sea by a narrow elongate strip of land

(D) All of these

103. Fissile claystone is called :

(A) Mudstone

(B) Siltstone

(C) Marl

(D) Shale

104. The difference between conglomerate and breccias is / are :

(A) Conglomerate is composed of angular grains but breccias are made up of rounded grains

(B) Conglomerate is fissile but breccias is hard

(C) Conglomerate and breccia are composed of round and angular grains respectively.

(D) None of these

105. Identify the sequence comparing chemical composition vs. rock name :

Chemical Composition		Rock name
(a)	$CaCO_3$	(i) Limonite
(b)	$FeO(OH).nH_2O$	(ii) Halite
(c)	$CaSO_4.2H_2O$	(iii) Dolostone
(d)	$CaMg(CO_3)_2$	(iv) Limestone
(e)	NaCl	(v) Rock Gypsum

Identify the correct sequence :

(A) (a) (iv), (b) (ii), (c) (v), (d) (c), (e) (i)

(B) (a) (iv), (b) (i), (c) (v), (d) (iii), (e) (ii)

(C) (a) (iii), (b) (ii), (c) (v), (d) (i), (e) (iv)

(D) (a) (iv), (b) (ii), (c) (iii), (d) (i), (e) (v)

106. Examples of high grade schist is :

(A) Garnet-schist

(B) Chlorite-schist

(C) Mica-schist

(D) Talc-schist

107. Gneisses formed from sedimentary rocks is called :

(A) Augen gneiss

(B) Banded gneiss

(C) Paragneiss

(D) Orthogneiss

108. In most metamorphic belts the lowest-grade metamorphism encountered is :

(A) Epidote-amphibolite facies

(B) Greenschist facies

(C) Zeolite facies

(D) Eclogite facies

109. In thermodynamics, an important variable is free energy (G), The free energy for a reaction is defined as :

(A) $\Delta G = \Delta H - T \Delta S$

(B) $\Delta G = - SdT + VdP$

(C) $d \Delta G = - \Delta SdT + \Delta VdP$

(D) All of these

Where, Δ = the difference in the thermodynamic function between the products and reactants, H = enthalpy, V = volume, and S = entropy.

110. Serpentine is a mineral group that includes :

(A) Antigorite

(B) Chrysotile

(C) Lizardite

(D) All of these

111. Tilley (1925) modified Barrow's originally defined metamorphic zones. These zones were defined in terms of the first appearance of key minerals with increasing grade, the zones are :

(A) Chlorite-biotite-garnet-staurolite-kyanite-sillimanite

(B) Chlorite-biotite-garnet-staurolite-kyanite-sillimanite

(C) Biotite-chlorite-staurolite-garnet-kyanite-sillimanite

(D) None of these

112. Wollastonite ($CaSiO_3$) forms in quartz-excess rocks by the reaction :

(A) $CaMg(CO_3)_2 + SiO_2 = MgO + CaSiO_3 + CO_2$

(B) $CaCO_3 + MgO + SiO_2 = CaSiO_3 + MgO + CO_2$

(C) $CaCO_3 + SiO_2 = CaSiO_3 + CO_2$

(D) All of these

113. The uppermost layer in an ophiolite is composed of _____ mostly pelagic mud, although chert may be common in some places :

(A) Alluvial sediments

(B) Evaporites

(C) Deep-water sediments

(D) None of these

With increasing pressure during metamorphism, actinolite

114. Disappears as the major amphibole and a sodic amphibole appears, consequent to this the rock turns blue, hence the rock is named _____.

(A) Amphibole schist

(B) Actinolite-amphibole schist

(C) Blue schist

(D) None of these

115. Three metamorphic facies contain amphibole as a critical phase. Find out the correct option :

(A) In greenschist (or the albite-epidote hornfels) facies the amphibole is actinolite

(B) In amphibolite (or hornblende hornfels) facies the amphibole is hornblende

(C) In blueschist facies the amphibole is sodic amphibole

(D) All of these

116. Name the mineral formed during metamorphism of the following :

$SiO_2 + NaAlSi_2O_6.H_2O =$ _____ $+ H_2O$

Quartz analcime

(A) Orthoclase ($KAlSi_3O_8$)

(B) Sanidine [K,Na)$AlSi_3O_8$]

(C) Microcline (triclinic)[19] $KAlSi_3O_8$

(D) Albite ($NaAlSi_3O_8$)

117. The last isogradic reaction forming coexistent sillimanite + biotite + garnet involves breakdown of staurolite : The reaction involved :

Staurolite + muscovite + quartz = sillimanite + biotite + _____ + water.

(A) Garnet

(B) Kyanite

(C) Amphibole

(D) K-feldspar

118. Thermal metamorphism of argillaceous rocks like shale causing large sized crystals (porphyroblasts) set in a fine grained rock, the texture/ structure is known as :

(A) Porphyry texture

(B) Granoblastic texture

(C) Maculose structure

(D) Poikilotopic structure

119. The high-energy strained grains may recrystallize by _____ and subsequent growth of completely new, _____, strain free grains.

(A) Nucleation, lowest-energy

(B) Melting, lowest-energy

(C) Solid solution, melting

(D) None of these

120. What sequence it follows when shales pass through successive stages of higher grades of metamorphism ?

(A) Shale, phyllite, gneiss, schist, slate

(B) Shale, phyllite, slate, schist, gneiss

(C) Shale, slate, phyllite, gneiss, schist

(D) Shale, phyllite, slate, gneiss, schist

121. The capacity of a stone to withstand bending load is termed as :

(A) Shear strength

(B) Bending strength

(C) Lateral strength

(D) Transverse strength

122. Which rock possesses high compressive strength ?

(A) Metamorphic rock

(B) Coarse grained sedimentary rock

(C) Igneous rocks

(D) Fine grained metamorphic rocks

123. What is the maximum force expressed per unit area, which a stone can withstand without rupturing ?

(A) Shear strength

(B) Tensile strength

(C) Compressive strength

(D) Bending strength

124. The density of the rock with natural moisture content is :

(A) Dry density

(B) Bulk density

(C) Wet density

(D) Natural density

125. What is Q-system in geology ?

(A) A classification system for soil and rock masses with respect to stability of underground openings

(B) A classification system for soil masses with respect to stability of underground openings

(C) A classification system for rock masses with respect to stability of underground openings

(D) All of these

126. Rock Quality Designation (RQD) refers to which property of rocks ?

(A) Degree of jointing or fracture in a rock mass

(B) Rock Mass Rating

(C) Q-system

(D) Strength Index

127. Hydraulic Conductivity, K(cm/s, m/s), a measure of the ability of a rock or soil to transmit water. The relationship is defined by the equation _____.

Where, k = intrinsic permeability ρ = density of water μ = viscosity of water g = acceleration due to gravity :

(A) $K = k^*/[(\rho g)^*\mu]$

(B) $K = k^*[(\rho)/\mu g]$

(C) $K = k^*[(\rho g)/\mu]$

(D) None of these

128. The index property of stones considered as an important type of Compressive strength _____.

(A) Drained

(B) Undrained

(C) Confined

(D) Unconfined

129. Ratio between the total pore spaces of rocks and the total volume of the rocks expressed in percentage is its _____.

(A) Pore ratio

(B) Bulk pore ratio

(C) Porosity

(D) Void ratio

130. The California Bearing Ratio (CBR) is a _____ for evaluation of the mechnical strength of road subgrades and base curves.

(A) Direct Shear Test

(B) Penetration test

(C) Slake Durability Index Test

(D) Proctor Compaction Test

131. A highly focuses beam of light is used in which sensors ?

(A) Ground penetrating radar

(B) Side looking radar

(C) Lidar

(D) Sonar

132. What would be the computer memory for a 512 × 512 image of 8 bits ?

(A) 8 MB

(B) 5 MB

(C) 512 MB

(D) 2.1 MB

133. A geostationary satellite is placed above the Earth's surface at an altitude of about _____.

(A) 20,000 km

(B) 35,000 km

(C) 36,000 km

(D) 45,000 km

134. The basic requirements of any sensor system is _____.

(A) Radiometric resolution

(B) Spatial resolution

(C) Spectral resolution

(D) All of these

135. Pick up the appropriate frequency regions of Sun's radiation :

(A) Infrared frequency region

(B) Radio frequency region

(C) Both (A) and (B)

(D) None of these

136. Specular reflection is the reflectance from a surface if it follows :

(A) Lambert's cosine law

(B) Planktan's law

(C) Snell's law

(D) All of these

137. A hot object emits thermal energy in the form of :

(A) Waves

(B) Rays

(C) Energy

(D) Radiations

138. The arrangement of terrain features which provides attributes the shape, size and texture of objects, is called :

(A) Temporal variation

(B) Spectral variation

(C) Spatial variation

(D) All of these

139. The process of systematically subdividing a geographic domain using Proximal region is known as :

(A) Dirichlet tessellation

(B) Thiessen polygon

(C) Voronoi diagram

(D) Nearest Neighbourhood regions

140. Which of the following resampling algorithm retains the raw pixel values from the original image ?

(A) Cubic covolution

(B) Bilinear

(C) Nearest neighbour

(D) Spline

141. GIS represents unrelated information of location using _____.

(A) Key

(B) Key index

(C) Key index variable

(D) Spatial index

142. Name the missing component of a working GIS :

(A) Hardware and software

(B) Data

(C) People and methods

(D) All of these

143. Which of the following doesn't indicate a topology error ?

(A) Gaps between polygons

(B) Silver polygons

(C) Polygonal features

(D) Unclosed polygons

144. GIS output uses which of the following format ?

(A) HTML

(B) GIF

(C) PDF

(D) None of these

145. The First Law of Geography states, "everything is related to everything else, but near things are more related than distant things" is known as :

(A) Anthony's first law of Geography

(B) Kepler's first law of Geography

(C) Tobler's first law of Geography

(D) Thompson's first law of Geography

146. Identify the server based hardware platform of GIS :

(A) STAAD Pro

(B) Google-maps

(C) Arc GIS

(D) Autodesk Revit

147. Database management system incorporates which of the following :

(A) Hierarchical database systems

(B) Network database systems

(C) Object-oriented database systems

(D) All of these

148. Which of the following is true about a full-fledged GIS package ?

(A) Data capture and preparation

(B) Data storage and data analysis

(C) Presentation of spatial data

(D) All of these

149. How to find or express a location on a Cadastral Map ?

(A) UTM Grid Coordinate

(B) With metric units

(C) Hand Level

(D) Using planimeter

150. Which is the most suitable scale for Cadastral mapping ?

(A) 1 : 50 to 1 : 100

(B) 1 : 100 to 1 : 400

(C) 1 : 500 to 10,000

(D) 1 : 25,000 to 1 : 50,000

SPACE FOR ROUGH WORK

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